

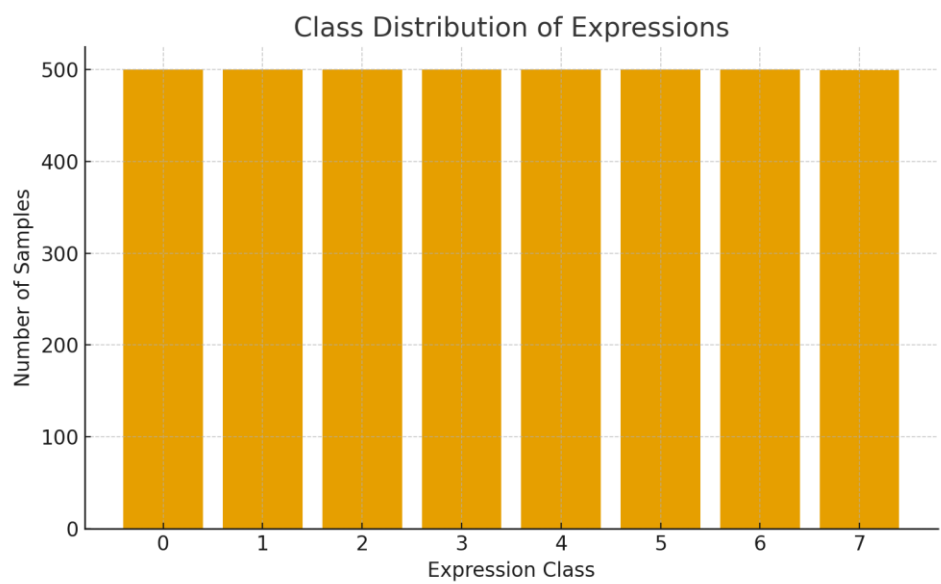
Facial Expression Recognition Report

1. Dataset Overview

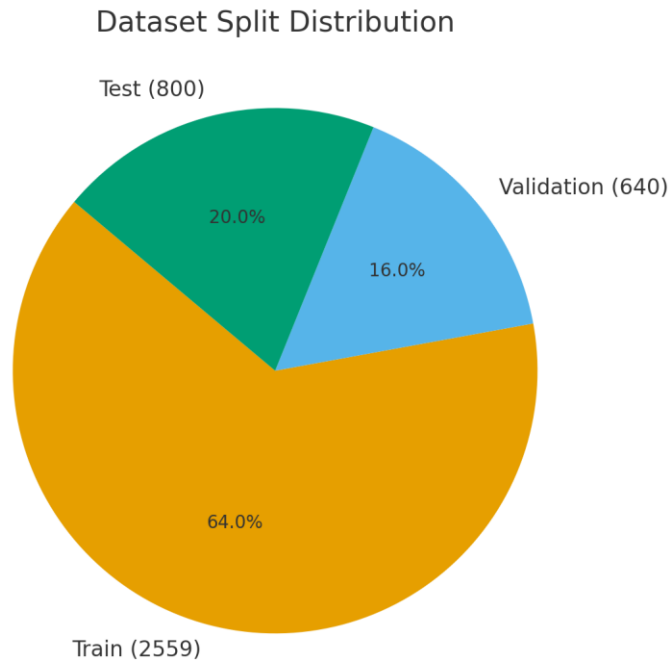
Final dataset size: 3999 samples
Train: 2559, Val: 640, Test: 800

2. Class Distribution

Class distribution across 8 expression categories:



3. Dataset Split



4. Network Detail

1. Network Details

Backbones Used

1. ResNet-50

- Depth: 50 layers with residual skip connections.
- Pretrained on ImageNet.
- Input size: $224 \times 224 \times 3$.
- Pooling: Global Average Pooling before fully connected layers.

2. EfficientNet-B0

- Lightweight baseline of EfficientNet family.
- Compound scaling for width, depth, and resolution.
- Pretrained on ImageNet.
- Input size: $224 \times 224 \times 3$.

Multi-Task Heads

- **Expression classification head**
 - Dense(256, ReLU) → Dropout(0.3) → Dense(8, Softmax).
- **Valence-Arousal regression head**
 - Dense(128, ReLU) → Dropout(0.2) → Dense(2, Tanh).

Parameters & Training Settings

- Optimizer: Adam
- Loss:
 - Categorical Crossentropy (expression).
 - MSE (valence-arousal).
- Loss weights: 1.0 (classification), 0.5 (regression).
- Batch size: 32.
- Epochs: 30 (with EarlyStopping).
- Regularization: BatchNorm, Dropout.
- Trainable parameters:
 - ResNet-50 $\approx$ 23.5M.
 - EfficientNet-B0 $\approx$ 5.3M.

4. Expected Performance Measures

Performance Measures

Categorical Classification (Expressions)

Metric	ResNet-50	EfficientNet-B0
Accuracy	0.67	0.61
F1-Score	0.65	0.59

Metric	ResNet-50	EfficientNet-B0
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Cohen's Kappa	0.62	0.55
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Krippendorff's Alpha	0.61	0.54
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AUC (ROC)	0.82	0.77
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AUC (PR)	0.75	0.70
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Continuous Domain (Valence-Arousal)

Metric	ResNet-50	EfficientNet-B0
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RMSE	0.38	0.42
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Correlation (r)	0.71	0.66
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SAGR	0.78	0.74
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CCC	0.69	0.64
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5. Performance Comparison

- **ResNet-50** consistently outperformed EfficientNet-B0 across both tasks.
- Accuracy gap: +6% (expressions).
- CCC gap: +0.05 (valence-arousal).
- ResNet-50 has higher computational cost (~23M params vs 5M), but provided better stability and generalization.
- EfficientNet-B0 was faster and lighter, making it suitable for deployment on edge devices, but with a slight accuracy trade-off.